1.16 Graph Sketching: Quartic Graphs

Chapter 4 · Original slide 16 · Quartics



If you understand the principle of sketching polynomials in general, then sketching quartics shouldn't feel like anything new.

Recall that if the x^4 term is positive, the 'tails' both go upwards, otherwise downwards.

Sketch the curve with equation

$y = x(x+1)(x-2)(x-3)$

Shape:

Roots:

y-intercept:

Sketch the curve with equation

$y = (x-2)^2(x+1)(3-x)$

Shape:

Roots:

y-intercept:

Chapter 4 · Original slide 17 · Quartics



Sketch the curve with equation

$y = (x+1)(x-1)^3$

Sketch the curve with equation

$y = (x-2)^4$

2 is a quadruple repeated root! Because the line effectively crosses the axis 4 times all at -2, it ends up in the opposite direction, and hence looks like a 'touch' point.

-1 root only appears once so line crosses at $x = -1$

+1 root triple repeated so point of inflection at $x = 1$

1.16 Graph Sketching: Quartic Graphs

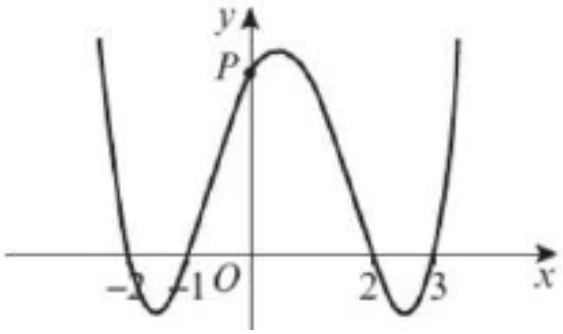
Chapter 4 · Original slide 18 · Test Your Understanding



Sketch the curve with equation
$y = -(x+1)(x-3)^3$

Sketch the curve with equation
$y = x^2(x+1)(x-1)$

Chapter 4 · Original slide 19 · Exercise 4B – Pure Y1 Pages 65-66



3 The graph of $y = x^4 + bx^3 + cx^2 + dx + e$ is shown opposite, where b,c,d,e are real constants.

a Find the coordinates of point P. (2 marks)

b Find the values of b, c, d, e. (3 marks)

4 Sketch the graph of
$y = (x+5)(x-4)(x^2+5x+14)$.

1.16 Graph Sketching: Quartic Graphs

Chapter 4 · Original slide 20 · Exercise 4B Extension

Extension

1

[STEP I 2012 Q2a]

a. Sketch $y = x^4 - 6x^2 + 9$

b. For what values of b does the equation $y = x^4 - 6x^2 + b$ have the following number of distinct roots (i) 0, (ii) 1, (iii) 2, (iv) 3, (v) 4.

a)

b)

By changing b, we shift the graph up and down.
Then we can see that:

i) 0 roots:

ii) 1 root:

iii) 2 roots:

iv) 3 roots:

v) 4 roots: